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Josephson Effect

[ALC - B24]

M766

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Experiment Date: 27.05.2025

Submission Date: 13.06.2025

Abstract

This report presents an experimental investigation of the Josephson effect, a macroscopic quantum phenomenon observed when two superconductors are weakly coupled through a thin barrier. Using a point-contact Josephson junction formed from niobium wires and cooled below its critical temperature of 9.3 K, both DC and AC Josephson effects were studied. The DC effect was characterized by a supercurrent flowing without an applied voltage, while the AC effect, triggered by a constant voltage, produced electromagnetic radiation in the microwave range.

Key theoretical concepts including Cooper pair formation, flux quantization, and the Meissner effect, were explored and connected to the observed electrical behavior. The RSJ (Resistively Shunted Junction) model was applied to describe the nonlinear dynamics of the junction, allowing the deduction of the governing differential equation for the superconducting phase difference. The influence of magnetic fields and the McCumber parameter on the I–V characteristics was also examined.

Experimental measurements included hysteretic and non-hysteretic current–voltage curves, as well as detection of microwave emission harmonics. These results validate the theoretical predictions of the Josephson effect and its dependence on parameters such as capacitance, resistance, and external magnetic flux. The study demonstrates the fundamental role of quantum coherence in superconducting systems and highlights the precision of Josephson junctions for voltage and magnetic field sensing applications.

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1 Introduction

The Josephson effect is a fascinating example of how quantum mechanics can influence macroscopic systems in ways that are directly measurable. First predicted by Brian D. Josephson in the early 1960s, the effect describes the tunneling of Cooper pairs—paired electrons—from one superconductor to another across a thin insulating barrier. This quantum tunneling results in a supercurrent that can flow even in the absence of an applied voltage. When a voltage is applied, the system emits radiation in the microwave frequency range [1].

This remarkable behavior has found important applications, particularly in the field of high-precision measurements. For instance, Superconducting Quantum Interference Devices (SQUIDs) leverage the Josephson effect to detect minute magnetic fields—down to the level of a single magnetic flux quantum [8]. Josephson junctions are also employed as ultra-stable voltage standards and in fast-switching superconducting electronics.

In this experiment, we aim to understand the Josephson effect in a practical setting by studying a point contact between two niobium superconductors. By adjusting the physical distance and temperature, we are able to examine both the DC and AC Josephson effects. We record and analyze current-voltage characteristics and detect the associated electromagnetic radiation.

These experimental results allow us to determine essential parameters such as the **McCumber parameter**, the **critical current**, and the fundamental constant ratio $\frac{2e}{h}$, which connects voltage and frequency in Josephson systems.

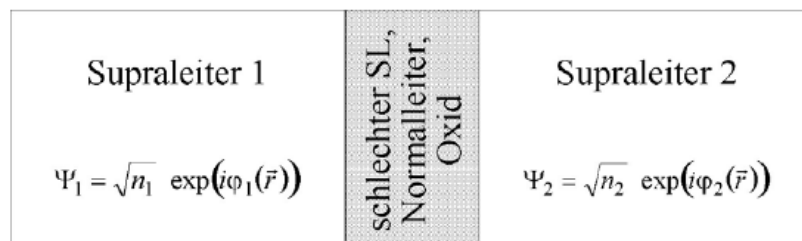


Figure 1.1: Schematic representation of a Josephson contact via a weak link

2 Theoretical Background

Before performing the experiment, it is essential to understand key concepts related to superconductivity and the Josephson effect.

2.1 Critical Temperature

The **critical temperature**, denoted as T_c , is the threshold below which a material transitions into the superconducting state. Above T_c , the material behaves like a conventional conductor where electrical resistance arises due to electron scattering from phonons and impurities. However, when cooled below T_c , the resistance abruptly drops to zero [4].

This dramatic change reflects a fundamental shift in the material's quantum state. Electrons form bound pairs, known as *Cooper pairs*, that move coherently through the lattice without scattering. This collective motion enables the flow of current without resistance.

In addition to the disappearance of resistance, superconductors also exhibit perfect diamagnetism below T_c , expelling magnetic fields from their interior—a phenomenon known as the *Meissner effect*. Thermodynamic properties such as specific heat also show discontinuities at the transition, confirming its nature as a true phase change [4].

In our experiment, the superconductor used is **niobium**, which has a critical temperature of approximately 9.3 K. This means all Josephson-related phenomena—such as Cooper pair tunneling and microwave emission—can be reliably observed only when the system is cooled well below this temperature using liquid helium.

2.2 Cooper Pairs

One of the cornerstones of superconductivity is the formation of **Cooper pairs**—bound states of two electrons that move through a superconductor without resistance. In a normal conductor, electrons scatter from lattice vibrations and impurities, leading to resistive losses. However, below the critical temperature, an effective attractive interaction mediated by phonons enables electrons to form weakly bound pairs [8].

This phenomenon, introduced by Leon Cooper in 1956 and developed into the BCS theory by Bardeen, Cooper, and Schrieffer in 1957 [3], describes how an electron moving through the lattice distorts it locally, attracting another electron to this positively charged region. The result is a pair of electrons that move in a correlated fashion despite their like charges [7].

Unlike classical particles, Cooper pairs behave as extended quantum wave packets and obey *Bose–Einstein statistics*. This allows many pairs to occupy the same ground state, flowing without scattering or resistance [1]. The coherence of these pairs underpins key superconducting phenomena such as flux quantization and the Josephson effects [6].

In Josephson junctions, Cooper pairs tunnel through weak links, enabling supercurrents to flow and generating microwave radiation under a voltage bias. Their coherent nature is central to interpreting the observations in this lab.

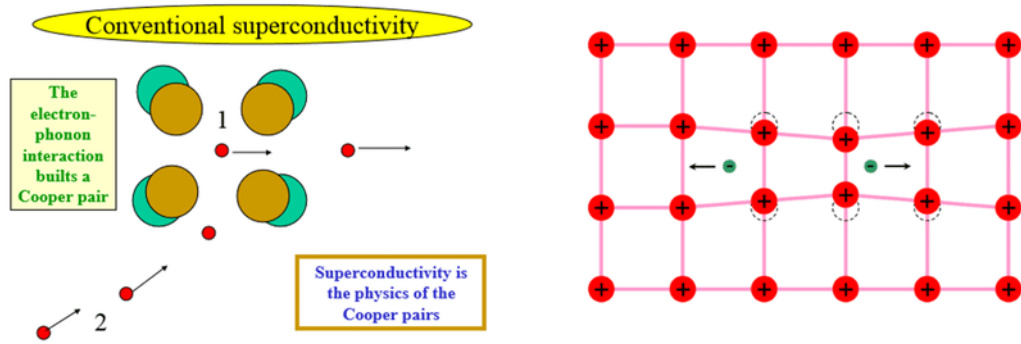


Figure 2.1: Illustration of Cooper pair formation via electron-phonon interaction. Adapted from [5].

2.3 Meissner Effect

One of the most remarkable features of superconductors is their ability to expel magnetic fields from their interior once cooled below a critical temperature. This phenomenon, known as the **Meissner–Ochsenfeld effect**, was first observed in 1933 by Walther Meissner and Robert Ochsenfeld. Unlike a perfect conductor that merely preserves the magnetic field during cooling, a superconductor *actively* expels magnetic flux, irrespective of its magnetic history. This behavior marks a true phase transition into a thermodynamically distinct state [8, 3].

To illustrate this, consider a perfect conductor with zero resistance, $R = 0$. From Faraday’s law for a loop enclosing area A , the induced voltage U is:

$$U = \oint \vec{E} \cdot d\vec{s} = -\frac{d\vec{B}}{dt} \cdot \vec{A}$$

If $R = 0$, then by Ohm's law $IR = 0$, implying $U = 0$, and thus $\frac{d\vec{B}}{dt} = 0$. This means the magnetic flux remains constant, and the conductor "remembers" the initial field. In contrast, a superconductor cooled below T_c always becomes flux-free, exhibiting the Meissner effect [8].

The theoretical basis was provided by the London brothers through two phenomenological equations. Starting from Newton's law for a superconducting electron of mass m and charge $-e$:

$$m \frac{d\vec{v}}{dt} = -e\vec{E}$$

Introducing the superconducting current density $\vec{j}_s = -en_s\vec{v}$, where n_s is the density of superconducting carriers, yields the **first London equation**:

$$\frac{d\vec{j}_s}{dt} = \frac{n_se^2}{m}\vec{E}$$

Defining the London constant $\Lambda_L = \frac{m}{n_se^2}$, this becomes:

$$\frac{d\vec{j}_s}{dt} = \frac{1}{\Lambda_L}\vec{E}$$

Using Maxwell's equation $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ and taking the time derivative of the curl of $\vec{j}_s$ gives:

$$\frac{\partial}{\partial t} (\Lambda_L \nabla \times \vec{j}_s + \vec{B}) = 0$$

Assuming the expression inside the derivative is constant and equal to zero, we arrive at the **second London equation**:

$$\nabla \times \vec{j}_s = -\frac{1}{\Lambda_L}\vec{B}$$

Combining this with Ampère's law, $\nabla \times \vec{B} = \mu_0\vec{j}_s$, we get:

$$\nabla^2 \vec{B} = \frac{1}{\lambda_L^2} \vec{B}$$

where $\lambda_L = \sqrt{\frac{\Lambda_L}{\mu_0}}$ is the **London penetration depth**. Solving this in one dimension (e.g., along the z -axis), we find:

$$B_x(z) = B_x(0) \cdot e^{-z/\lambda_L}$$

This solution shows that the magnetic field decays exponentially within the superconductor, rather than penetrating fully. Typical values of λ_L in conventional superconductors range from 100 to 500 nm [7].

The Meissner effect, together with the London equations, confirms that superconductivity is not merely perfect conductivity, but a distinct macroscopic quantum state. This forms the basis for other effects such as *flux quantization* and the *Josephson effect*, both of which rely on coherent, phase-locked superconducting wavefunctions.

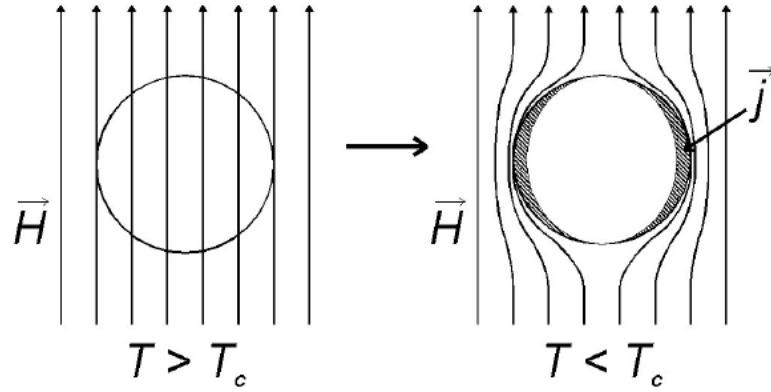


Figure 2.2: Illustration of the Meissner effect: expulsion of magnetic field from a superconductor below T_c

2.4 Flux Quantization

One of the hallmark signatures of the superconducting state is the quantization of magnetic flux. When a superconducting ring is exposed to a magnetic field, the total magnetic flux Φ passing through the ring does not vary continuously. Instead, it is constrained to discrete values—integer multiples of a fundamental unit of flux. This phenomenon is a direct consequence of the quantum mechanical nature of superconductivity [8, 3].

The starting point is the macroscopic wavefunction that describes Cooper pairs in a superconductor:

$$\Psi(\vec{r}) = \sqrt{n} \cdot \exp(i\phi(\vec{r}))$$

Here, n is the Cooper pair density and ϕ is the phase of the wavefunction. In the presence of a vector potential $\vec{A}$, the quantum mechanical current density is:

$$\vec{j} = \frac{iq}{2m_0} (\Psi \nabla \Psi^* - \Psi^* \nabla \Psi) - \frac{nq^2}{m_0} \vec{A}$$

Substituting $q = -2e$ and $m_0 = 2m$ for Cooper pairs simplifies this to:

$$\vec{j} = -\frac{ne}{m} \nabla \phi - \frac{2ne^2}{m} \vec{A}$$

To evaluate this around a superconducting ring, we integrate over a closed loop:

$$\oint \vec{j} \cdot d\vec{s} = -\frac{ne^2}{m} \oint \vec{A} \cdot d\vec{s} - \frac{ne}{m} \oint \nabla \phi \cdot d\vec{s}$$

The requirement that the wavefunction be single-valued implies that its phase must change by $2\pi N$ over a closed loop:

$$\oint \nabla \phi \cdot d\vec{s} = 2\pi N, \quad N \in \mathbb{Z}$$

In the superconducting state, due to the Meissner effect, the internal current $\vec{j}$ is zero. Therefore:

$$\oint \vec{A} \cdot d\vec{s} = \frac{h}{2e} N$$

Using Stokes' theorem, this loop integral becomes:

$$\oint \vec{A} \cdot d\vec{s} = \iint_F (\nabla \times \vec{A}) \cdot d\vec{F} = \iint_F \vec{B} \cdot d\vec{F} = \Phi$$

Thus, the total magnetic flux through the superconducting ring is quantized:

$$\Phi = N \cdot \frac{h}{2e}, \quad N \in \mathbb{Z}$$

This result demonstrates that magnetic flux in a superconducting loop can only exist in discrete units of $\Phi_0 = \frac{h}{2e}$, known as the *flux quantum*. The factor of $2e$ arises because Cooper pairs act as the charge carriers.

This theoretical prediction has been experimentally verified. Superconducting rings exhibit quantized steps in magnetic flux, and this effect underpins sensitive magnetic field detection technologies such as **SQUIDS** (Superconducting Quantum Interference Devices), capable of detecting even a single flux quantum.

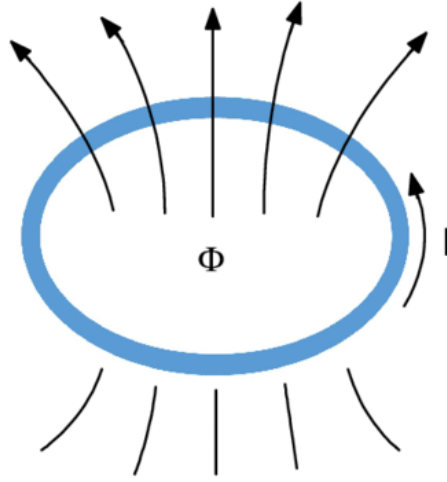


Figure 2.3: A superconducting ring with magnetic flux Φ threading through it, illustrating the condition for flux quantization. Adapted from [2].

2.5 Josephson Effect

When two superconductors are separated by a sufficiently thin barrier—such as an insulator or a weak link—they can form a Josephson junction. In this configuration, Cooper pairs can tunnel across the barrier due to quantum mechanical effects, even when no voltage is applied. This leads to a dissipationless current known as the *Josephson supercurrent*.

The supercurrent depends on the phase difference between the macroscopic wavefunctions of the superconductors on either side of the junction. This is described by the **first Josephson equation**:

$$I = I_c \sin(\gamma)$$

where I_c is the critical current, and γ is the gauge-invariant phase difference defined by:

$$\gamma = \phi_2 - \phi_1 + \frac{2e}{\hbar c} \int \vec{A} \cdot d\vec{s}$$

Here, ϕ_1 and ϕ_2 are the phases of the superconducting wavefunctions, and $\vec{A}$ is the magnetic vector potential, with $\vec{B} = \nabla \times \vec{A}$.

When the applied current exceeds I_c , tunneling of Cooper pairs alone is no longer sufficient. In this regime, quasiparticles contribute to the current, resulting in a voltage drop across the junction and a finite resistance R_n .

The presence of a voltage modifies the phase difference, as described by the **second Josephson equation**:

$$\frac{d\gamma}{dt} = \frac{2e}{\hbar} U$$

This implies that a constant voltage U leads to a linearly increasing phase difference $\gamma(t)$, which produces an alternating current through the junction at frequency:

$$f = \frac{2e}{h} U$$

Gauge Invariance: Both Josephson equations respect electromagnetic gauge invariance. Under a gauge transformation, the scalar potential φ and the vector potential $\vec{A}$ transform as:

$$\varphi \rightarrow \varphi - \frac{1}{c} \frac{d\chi}{dt}, \quad \vec{A} \rightarrow \vec{A} + \nabla\chi$$

To ensure observables such as current remain unchanged, the superconducting phase must transform as:

$$\phi \rightarrow \phi - \frac{2e}{\hbar c} \chi$$

This preserves the gauge-invariant phase difference γ and its time derivative:

$$\frac{d\gamma}{dt} = \frac{2e}{\hbar} U$$

confirming the consistency of the Josephson effect with gauge theory [3, 7].

2.6 Josephson Contacts and Tunnel Current

To experimentally observe the Josephson effect, two superconductors must be coupled in such a way that Cooper pairs can exchange quantum information across a thin non-superconducting barrier. These structures are known as *Josephson contacts* or *Josephson junctions*, and their behavior depends strongly on their physical construction [3].

Common types of Josephson contacts include:

- **Tunnel junction (oxide layer):** A very thin insulating layer (typically an oxide) separates the two superconductors. Cooper pairs can quantum tunnel through this barrier. These junctions are widely used in precision applications like voltage standards.
- **SNS junction (normal conductor):** A short segment of normal metal connects the superconductors. Due to the proximity effect, superconducting correlations extend into the normal region, enabling pair transport.
- **Point contact:** A sharp superconducting tip touches a flat superconducting surface. This type can behave as a tunnel junction or a metallic weak link depending on pressure and cleanliness.

In our experiment, we use a point contact between niobium wires. By adjusting the mechanical pressure between the tip and the base, we can tune the contact between tunnelling-like and metallic behaviour. This directly influences the current-voltage (I – V) characteristics: tunnel-like contacts can exhibit hysteresis, whereas metallic contacts are typically overdamped and non-hysteretic.

The current in a tunnel junction arises from quantum tunneling. While classical physics forbids electron flow through the barrier, quantum mechanics allows a finite probability for Cooper pairs to tunnel through. This enables a supercurrent to flow without voltage, provided the current remains below the critical threshold I_c .

Thus, Josephson contacts reveal both the macroscopic coherence of superconductivity and the essential quantum nature of transport across barriers.

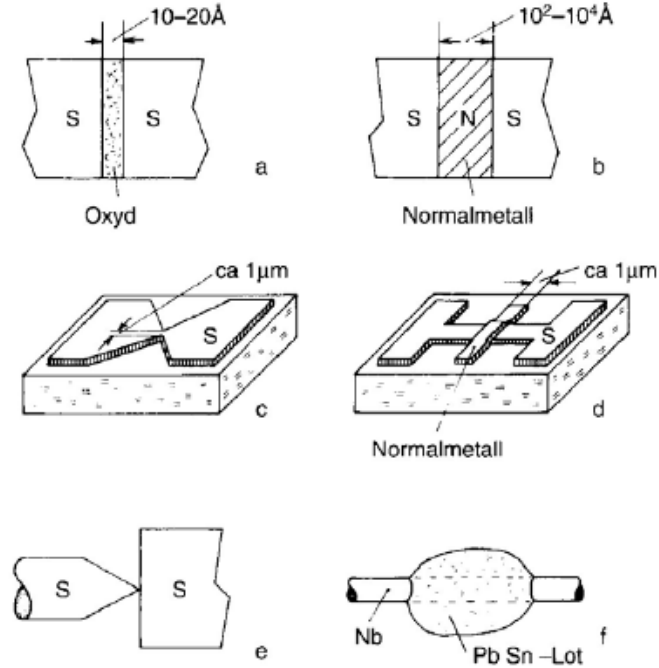


Figure 2.4: Different types of Josephson contacts: (a) oxide tunnel junction, (b) SNS junction, (c) constriction, (d) proximity-effect double layer, (e) point contact, (f) SLUG array. Adapted from [4].

2.7 DC and AC Josephson Effects

The Josephson effect manifests in two fundamental forms: the **DC Josephson effect** and the **AC Josephson effect**, depending on whether a voltage is applied across the junction.

DC Josephson Effect

In the absence of an applied voltage ($U = 0$), the phase difference γ across the Josephson junction remains constant:

$$\gamma = \gamma_0 = \text{const.}$$

From the first Josephson equation, this yields a constant supercurrent:

$$I = I_c \sin(\gamma)$$

The current can vary from $-I_c$ to $+I_c$, depending on the phase. This effect allows a supercurrent to flow with zero voltage—purely due to the coherent phase relationship between the superconductors.

AC Josephson Effect

When a constant voltage U is applied, the phase evolves in time according to the second Josephson equation:

$$\frac{d\gamma}{dt} = \frac{2e}{\hbar} U$$

Integrating over time gives:

$$\gamma(t) = \gamma_0 + \frac{2e}{\hbar} U t$$

Substituting into the first Josephson equation yields a time-dependent current:

$$I(t) = I_c \sin \left(\gamma_0 + \frac{2e}{\hbar} U t \right)$$

This describes an alternating current with frequency:

$$f = \frac{2e}{h} U$$

For instance, at 10 GHz:

$$U = \frac{fh}{2e} = \frac{(10 \times 10^9) \cdot (6.626 \times 10^{-34})}{2 \cdot 1.602 \times 10^{-19}} \approx 20.7 \mu\text{V}$$

The second harmonic at 5 GHz corresponds to approximately $10.35 \mu\text{V}$.

Effect of Magnetic Field

The Josephson current also depends on magnetic flux Φ through the junction due to its impact on the phase difference. The resulting critical current follows:

$$I_c(B) = I_c(0) \cdot \left| \text{sinc} \left(\frac{\pi \Phi}{\Phi_0} \right) \right|$$

Here, $\Phi = 2\lambda_L l B$, with λ_L as the London penetration depth, l the junction width, and $\Phi_0 = \frac{h}{2e}$ the flux quantum.

This pattern resembles the single-slit diffraction pattern and results from phase interference of the tunneling Cooper pairs in the magnetic field.

2.8 Current–Voltage Characteristics of Josephson Contacts

The current–voltage (I–V) behavior of a Josephson junction reflects the interplay of the Josephson effect, resistive current, and capacitive charging. The total current is:

- Josephson supercurrent: $I_c \sin \gamma$
- Resistive quasiparticle current: $\frac{U}{R_n}$
- Capacitive displacement current: $C \frac{dU}{dt}$

Using $\frac{d\gamma}{dt} = \frac{2e}{\hbar} U$, we obtain:

$$I_{\text{tot}} = I_c \sin \gamma + \frac{\hbar}{2eR_n} \frac{d\gamma}{dt} + \frac{\hbar C}{2e} \frac{d^2\gamma}{dt^2}$$

This resembles the equation for a damped driven harmonic oscillator:

$$F(t) = kx + c\dot{x} + m\ddot{x}$$

In this analogy:

- $I_{\text{tot}} \rightarrow$ external force
- $\gamma \rightarrow$ displacement
- $R_n \rightarrow$ damping
- $C \rightarrow$ mass (inertial term)

Such non-linear dynamics can generate harmonics of the AC Josephson frequency.

McCumber Parameter The I–V curve behavior is characterized by the **McCumber parameter**:

$$\beta_c = \frac{2\pi C R_n^2 I_c}{\Phi_0}$$

where $\Phi_0 = \frac{h}{2e}$.

1. **Underdamped regime** ($\beta_c > 1$): Exhibits hysteresis. The junction switches from super-current to resistive state at I_c , and back at a smaller retrapping current I_R :

$$\frac{I_R}{I_c} = \frac{4}{\pi\sqrt{\beta_c}}$$

2. **Overdamped regime** ($\beta_c < 1$): No hysteresis. For $\beta_c = 0$, the voltage is:

$$\langle U(t) \rangle = \pm R_n \sqrt{I_{\text{tot}}^2 - I_c^2}$$

This analysis helps classify junctions based on their I–V curves and distinguish between tunnel-type and metallic weak-link junctions.

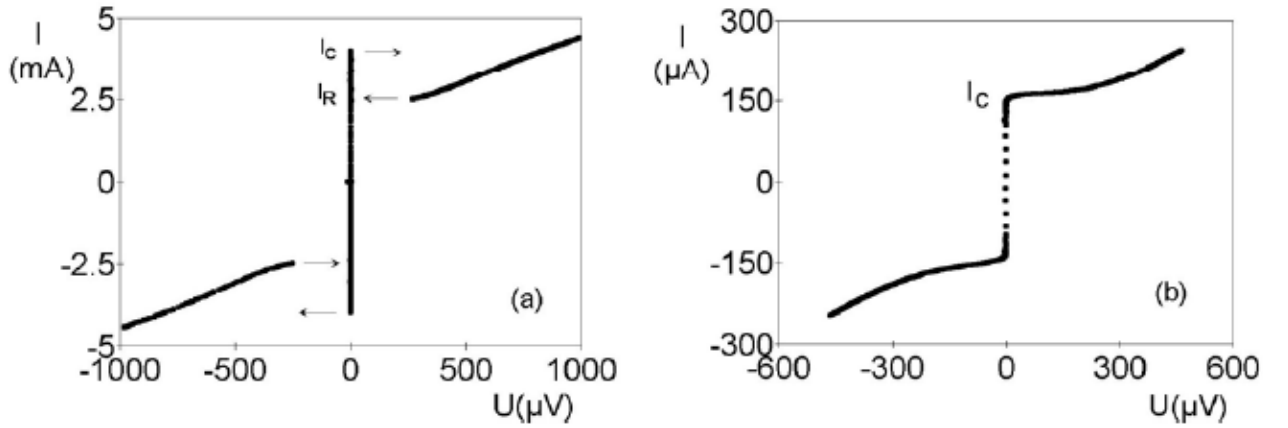


Figure 2.5: (a) Hysteretic I–V characteristic of an underdamped Josephson junction. (b) Non-hysteretic I–V curve of an overdamped junction. Adapted from [4].

2.9 Questions for Preparation

1. Why does the 4-point method of resistance measurements have to be used?

The 4-point method improves accuracy by eliminating the influence of wire and contact resistances, which is especially important in low-resistance setups like superconducting junctions. In this method, one pair of leads carries current, and a separate pair measures voltage. Because negligible current flows through the voltage leads, the voltage drop across the contact resistances does not affect the measurement. This ensures we measure only the intrinsic properties of the Josephson contact.

2. At which voltage are you expecting the radiation emission from the Josephson contact at the frequency of 10 GHz? At which voltage are you expecting the second harmonic for the same emission frequency?

The AC Josephson relation links voltage and frequency:

$$f = \frac{2e}{h}U \quad \Rightarrow \quad U = \frac{fh}{2e}$$

For $f = 10$ GHz:

$$U = \frac{(6.626 \times 10^{-34}) \cdot (10 \times 10^9)}{2 \cdot 1.602 \times 10^{-19}} \approx 20.7 \mu\text{V}$$

For the second harmonic ($f = 5$ GHz):

$$U \approx 10.35 \mu\text{V}$$

Answer: Emission occurs at $\sim 20.7 \mu\text{V}$ for 10 GHz and $\sim 10.35 \mu\text{V}$ for 5 GHz [3, 7].

3. Deduce the differential equation for γ in the RSJ model and the Josephson equations

In the RSJ model, the total current is:

$$I_{\text{tot}} = I_c \sin(\gamma) + \frac{U}{R_n} + C \frac{dU}{dt}$$

Using $\frac{d\gamma}{dt} = \frac{2e}{\hbar}U$, we substitute:

$$I_{\text{tot}} = I_c \sin(\gamma) + \frac{\hbar}{2eR_n} \frac{d\gamma}{dt} + \frac{\hbar C}{2e} \frac{d^2\gamma}{dt^2}$$

This nonlinear second-order equation resembles a driven damped harmonic oscillator [7]:

$$I_{\text{tot}} = I_c \sin(\gamma) + \alpha \frac{d\gamma}{dt} + \beta \frac{d^2\gamma}{dt^2}$$

with:

$$\alpha = \frac{\hbar}{2eR_n}, \quad \beta = \frac{\hbar C}{2e}$$

Josephson Equations:

$$I_s = I_c \sin(\gamma), \quad \frac{d\gamma}{dt} = \frac{2e}{\hbar}U$$

4. Which analogy exists between the parameters C , R_n , γ , and I of a Josephson contact and the parameters of a physical pendulum?

The RSJ model maps onto a driven damped pendulum [3]:

- $\gamma \leftrightarrow$ pendulum angle θ
- $I \leftrightarrow$ external torque $\tau(t)$
- $R_n \leftrightarrow$ damping coefficient $1/b$
- $C \leftrightarrow$ moment of inertia J

Both obey a similar differential equation:

$$I = I_c \sin(\gamma) + \alpha \frac{d\gamma}{dt} + \beta \frac{d^2\gamma}{dt^2} \quad \text{vs.} \quad \tau(t) = \tau_{\text{max}} \sin(\theta) + b\dot{\theta} + J\ddot{\theta}$$

This analogy helps to visualize quantum behavior with classical dynamics.

5. Show the transformation invariance of the second Josephson equation (Eq. 2.10)

The second Josephson equation:

$$\frac{d\gamma}{dt} = \frac{2e}{\hbar} U$$

To test gauge invariance, consider the transformations:

$$\varphi \rightarrow \varphi - \frac{1}{c} \frac{d\chi}{dt}, \quad \vec{A} \rightarrow \vec{A} + \nabla\chi, \quad \phi \rightarrow \phi - \frac{2e}{\hbar c} \chi$$

The gauge-invariant phase difference:

$$\gamma = \phi_2 - \phi_1 + \frac{2e}{\hbar c} \int \vec{A} \cdot d\vec{s}$$

After transformation:

$$\gamma' = \gamma, \quad \frac{d\gamma'}{dt} = \frac{d\gamma}{dt}$$

Similarly, $U' = U - \frac{1}{c} \frac{d}{dt}(\chi_2 - \chi_1)$, and the changes cancel out. Therefore, the second Josephson equation remains invariant:

$$\frac{d\gamma}{dt} = \frac{2e}{\hbar} U$$

Conclusion: The second Josephson equation is gauge invariant [3, 7].

3 Experimental Setup

3.1 Sample and Contact Preparation

The core component of our setup is a **niobium point contact**, which serves as the Josephson junction. This is established using a fine niobium tip brought into contact with a flat niobium wire. The entire configuration is mounted on a specialized assembly known as a **dipstick**, as illustrated in the schematic provided.

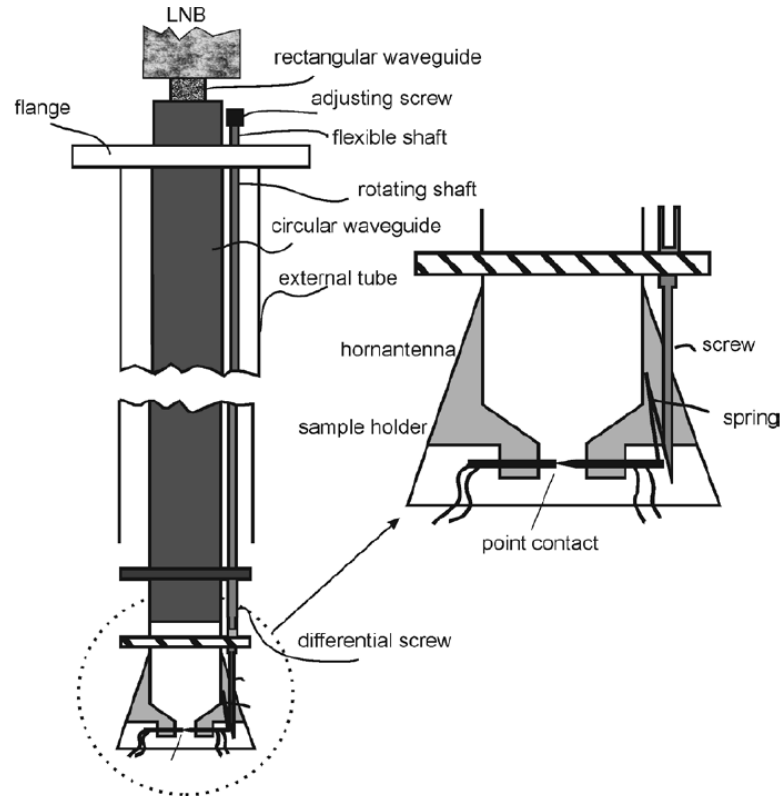


Figure 3.1: Dipstick for Josephson measurements

The precise method for shaping and attaching the niobium tip is vital for ensuring a clean and reproducible superconducting contact.

3.2 Dipstick Configuration

The dipstick is designed for insertion into a **liquid helium dewar**, enabling the entire system to cool below niobium’s critical temperature of approximately 9.3 K. This low temperature is essential for achieving superconductivity and observing the Josephson effects.

Integrated into the dipstick is a **circular waveguide** that operates in the 9.8 GHz to 11.5 GHz frequency range. A **horn antenna** is mounted at the base of this waveguide to capture electromagnetic radiation emitted by the Josephson contact. The contact is carefully positioned at the center of the antenna’s focus for optimal signal collection.

Mechanically, the dipstick allows the niobium tip to be **slowly and precisely moved** into contact with the flat niobium wire, forming the weak link required for the experiment.

3.3 Electrical Measurement Setup

The experiment utilizes a **four-point measurement technique** to obtain accurate current-voltage (I-V) characteristics of the Josephson junction while eliminating parasitic contact resistances.

- A programmable **DC current source** provides current up to ± 100 mA. The actual value of the applied current is monitored by **Digital Voltmeter 1 (DVM1)**. The voltmeter shows voltage, which must be converted to current using a conversion factor based on the current source’s selected range.
- The **voltage across the Josephson junction** is measured using **DVM2**.
- Both DVM1 and DVM2 are interfaced to a computer via the **GPIO bus**, and the data are acquired using the **kgPlot software**.

All electrical lines are equipped with **low-pass filters** to suppress high-frequency noise and interference.

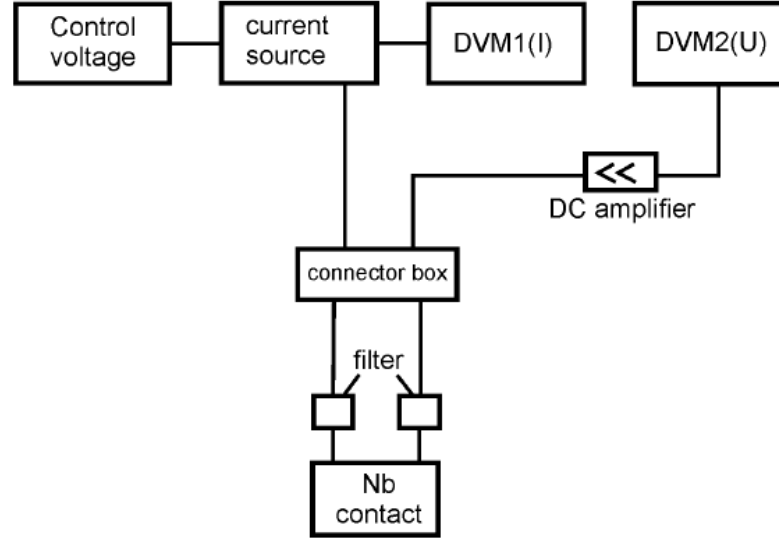


Figure 3.2: Diagram of the current and voltage measurements

3.4 Radiometer Configuration for Emission Detection

To measure the **microwave radiation** emitted from the Josephson contact (due to the **AC Josephson effect**), the setup includes a **radiometer system**.

- The horn antenna collects the emitted radiation and guides it via a waveguide to a **Low-Noise Block Downconverter (LNB)**.
- Inside the LNB, the signal is first amplified and then **mixed with a local oscillator signal** (around 10 GHz), resulting in an **intermediate frequency (IF)**.
- The IF signal is passed through a filter (either 30 MHz or 70 MHz window), then rectified by a diode, and finally measured by **Digital Voltmeter 3 (DVM3)**.
- The measured voltage is proportional to the **square root of the radiation power**.

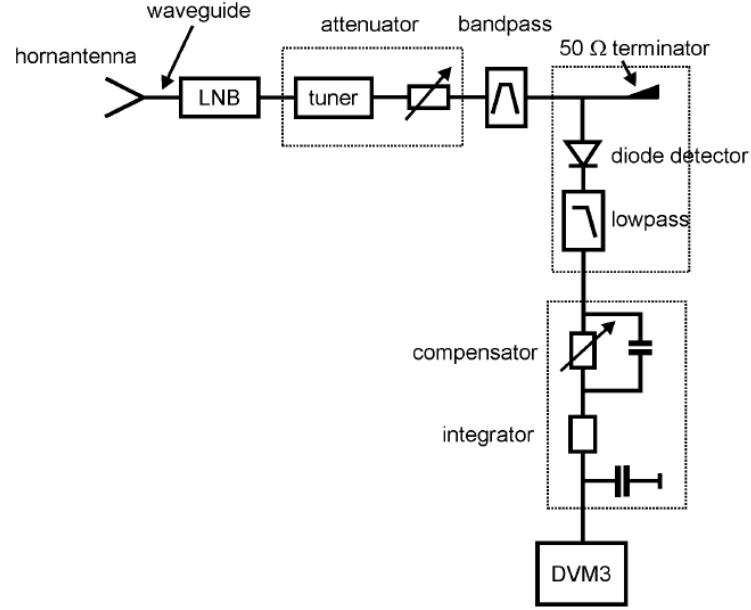


Figure 3.3: Radiometer set-up

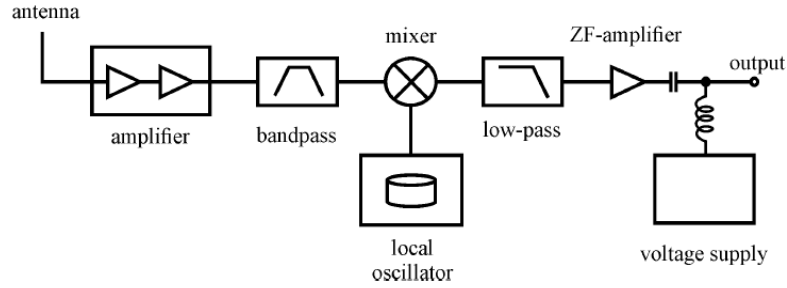


Figure 3.4: LNB set-up

This configuration enables precise detection of the first and second harmonic emission peaks.

3.5 Temperature Monitoring

A **PT100 platinum resistor**, placed near the sample, is used to continuously monitor the temperature during the experiment. The resistance measured is converted into temperature using a **calibration table**. Maintaining the temperature below 9.3 K is crucial for ensuring the system remains in the superconducting state.

4 Experimental Analysis

4.1 I-V Characteristics for Underdamped Condition

When the temperature was lowered, the junction entered the underdamped regime due to the increasing superconducting energy gap between the Cooper pair ground state and the excited states. This led to characteristic hysteresis behavior in the I-V curve. The junction abruptly switched from zero-voltage superconducting state to a finite voltage resistive state at the critical current I_c , and returned back at the retrapping current I_r during the backward sweep.

The extracted values for I_c and I_r were obtained by averaging multiple hysteresis loops. The results, including their uncertainties calculated by standard error propagation, are listed in Table 4.1.

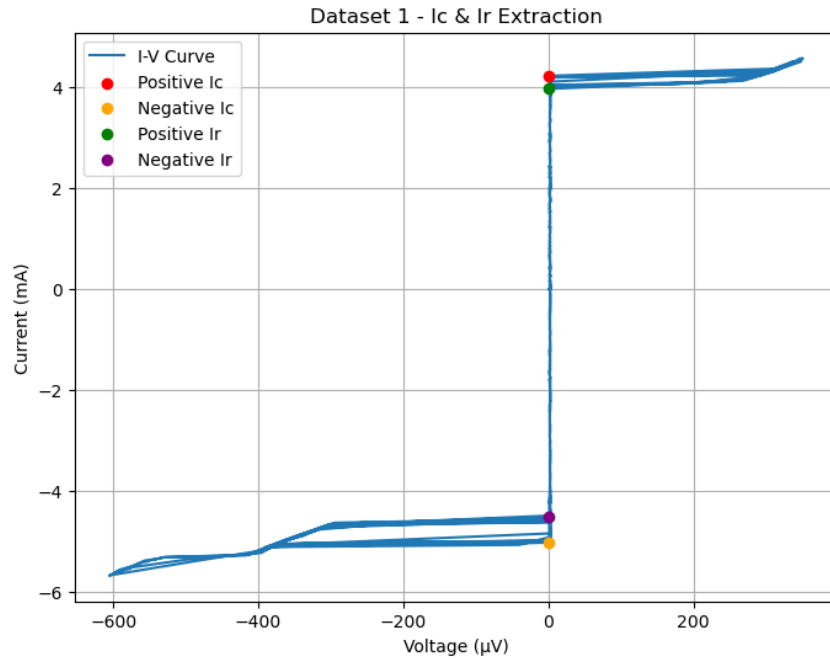


Figure 4.1: Measurement 1 I-V Curve Characteristics

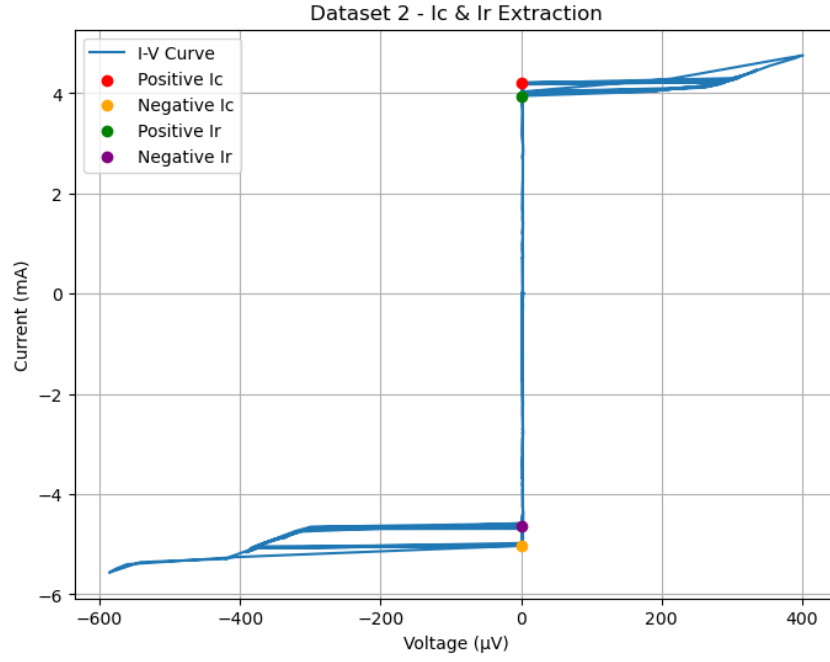


Figure 4.2: Measurement 2 I-V Curve Characteristics

Table 4.1: Critical and Retrapping Currents

Dataset	Branch	I_c (mA)	I_r (mA)
1	Positive	4.22 ± 0.02	3.96 ± 0.03
1	Negative	-5.02 ± 0.01	-4.52 ± 0.04
2	Positive	4.21 ± 0.02	3.95 ± 0.03
2	Negative	-4.64 ± 0.01	-4.64 ± 0.04

4.1.1 Calculation of McCumber Parameter

The McCumber parameter β_c was calculated according to:

$$\beta_c = \left(\frac{4I_c}{\pi I_r} \right)^2$$

The uncertainty in β_c was calculated using Gaussian error propagation:

$$\Delta\beta_c = \sqrt{\left(\frac{32I_c}{\pi I_r^2} \Delta I_c \right)^2 + \left(\frac{-32I_c^2}{\pi I_r^3} \Delta I_r \right)^2}$$

The resulting values are summarized in Table 4.2.

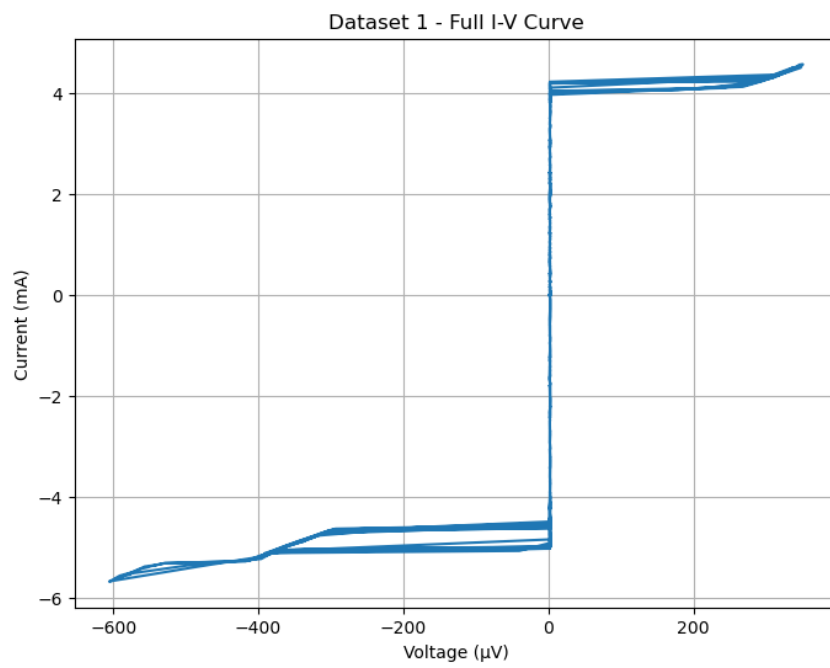


Figure 4.3: Measurement 1 I-V Curve with Hysteresis

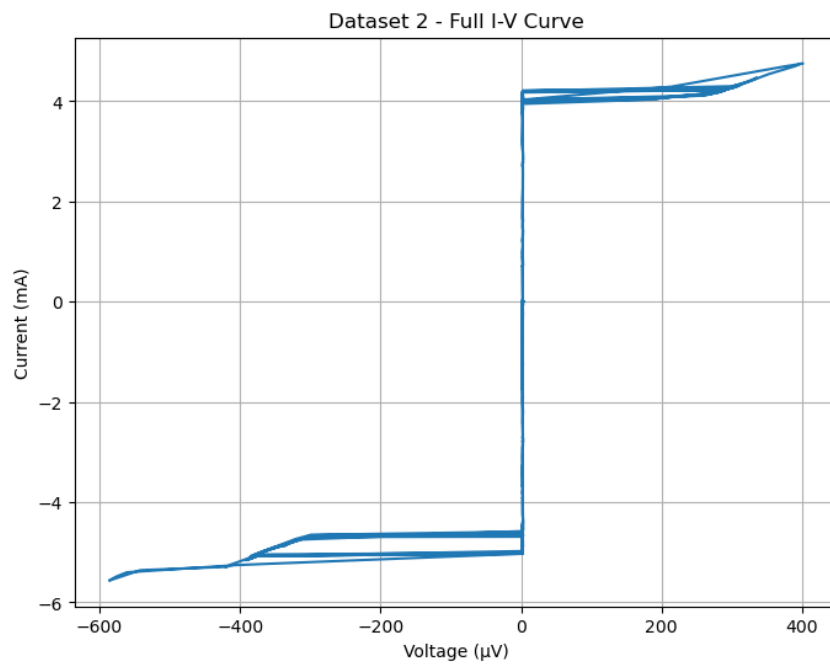


Figure 4.4: Measurement 2 I-V Curve with Hysteresis

Table 4.2: McCumber Parameters

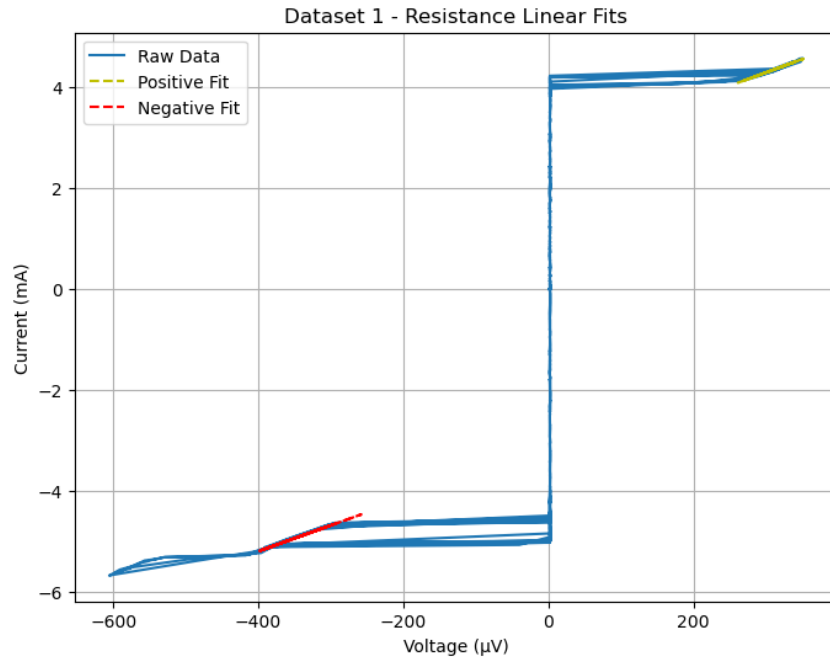
Dataset	Branch	β_c
1	Positive	1.84 ± 0.02
1	Negative	2.01 ± 0.018
2	Positive	1.85 ± 0.02
2	Negative	1.91 ± 0.02

4.1.2 Calculation of Normal Resistance

The normal state resistance R_n was extracted from linear fits of the resistive (ohmic) branches of the I-V curves. The slope of the fit gives the conductance G , and $R_n = 1/G$. The error was calculated as:

$$\Delta R_n = \frac{\Delta \text{slope}}{(\text{slope})^2}$$

The extracted resistances are listed in Table 4.3.

**Figure 4.5:** Measurement 1 I-V Curve with Resistance Linear Fits

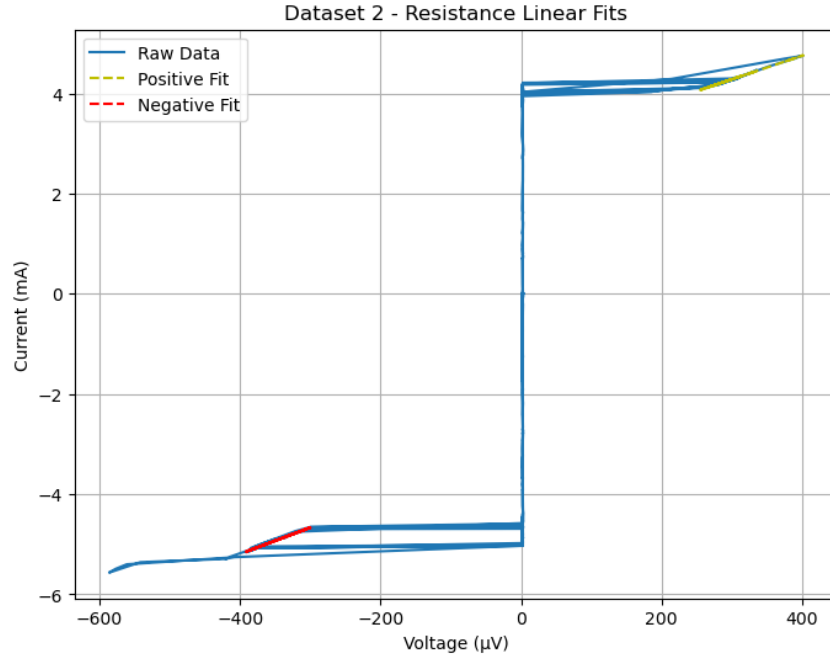


Figure 4.6: Measurement 2 I-V Curve with Resistance Linear Fits

Table 4.3: Normal Resistance from Linear Fits

Dataset	Branch	R_n (Ω)
1	Positive	0.189 ± 0.002
1	Negative	0.195 ± 0.003
2	Positive	0.212 ± 0.002
2	Negative	0.190 ± 0.002

4.1.3 Calculation of Junction Capacitance

The junction capacitance was finally calculated via:

$$C = \frac{\Phi_0 \beta_c}{2\pi R_n^2 I_c}$$

with $\Phi_0 = 2.067833848 \times 10^{-15}$ Wb. The uncertainty in C was calculated using Gaussian error propagation:

$$\Delta C = \sqrt{\left(\frac{\Phi_0 \beta_c}{2\pi R_n^2 I_c} \Delta \beta_c\right)^2 + \left(\frac{-\Phi_0 \beta_c}{\pi R_n^3 I_c} \Delta R_n\right)^2 + \left(\frac{-\Phi_0 \beta_c}{2\pi R_n^2 I_c^2} \Delta I_c\right)^2}$$

The final capacitance values are summarized in Table 4.4.

Table 4.4: Extracted Junction Capacitance

Dataset	Branch	C (pF)
1	Positive	3.97 ± 0.09
1	Negative	3.47 ± 0.11
2	Positive	3.20 ± 0.06
2	Negative	3.45 ± 0.09

4.2 Microwave Emission

4.2.1 Experimental Setup and Measurement Principle

A superconducting Josephson junction was employed to generate microwave emission due to the AC Josephson effect. To prevent interference with the Cooper pair oscillations, a non-hysteretic setup was used. A radiometer simultaneously measured the emitted microwave radiation and I-V characteristics at a target frequency of 11.3 GHz.

According to the AC Josephson relation:

$$U = \frac{fh}{2e}$$

the expected emission peak occurs near $U \approx \pm 23.4 \mu\text{V}$ for $f = 11.3$ GHz.

4.2.2 Critical Current and Differential Resistance at Emission Maxima

The critical current was determined from linear fits applied to both polarizations around the emission peak. The extracted values for both datasets are shown in Table 4.5.

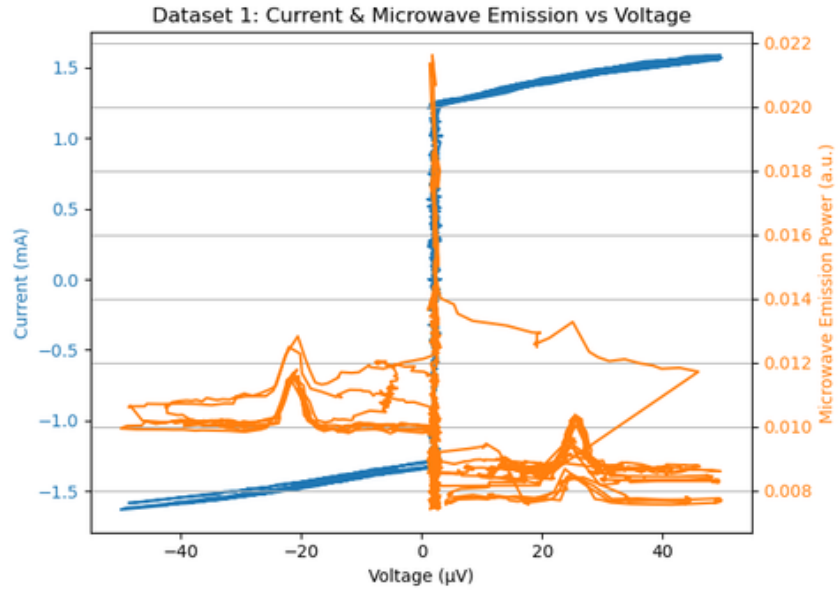


Figure 4.7: Measurement 1 I-V Curve with Microwave Emission

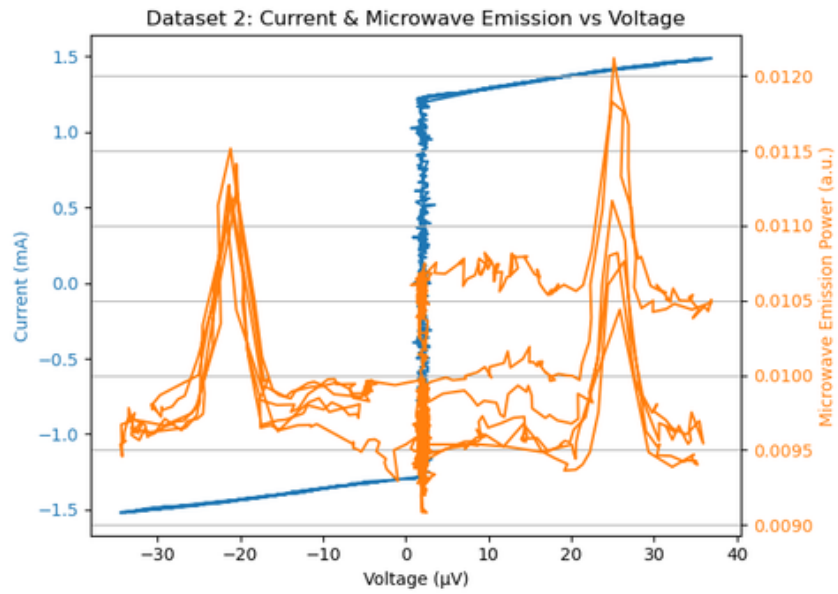


Figure 4.8: Measurement 1 I-V Curve with Microwave Emission

Table 4.5: Critical Current at Emission Maxima

Dataset	Branch	I_c (mA)
1	Positive	1.34 ± 0.03
1	Negative	-1.63 ± 2.33
2	Positive	1.29 ± 0.07
2	Negative	-1.52 ± 2.70

To calculate the differential resistance at the emission peak, Gaussian fits were applied to the emission power vs voltage data. The differential resistance R_d was extracted from the slope of the Gaussian fit at its maximum using:

$$y(x) = A \exp \left(-\frac{(x - \mu)^2}{\sigma^2} \right) + c$$

The extracted differential resistance values are shown in Table 4.6.

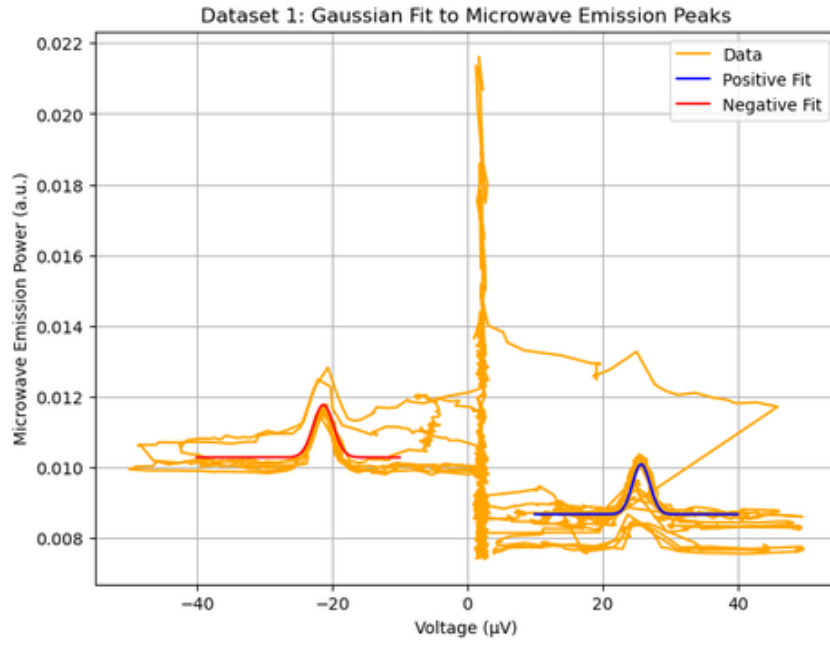


Figure 4.9: Gaussian Fits to Microwave Emission on Measurement 1

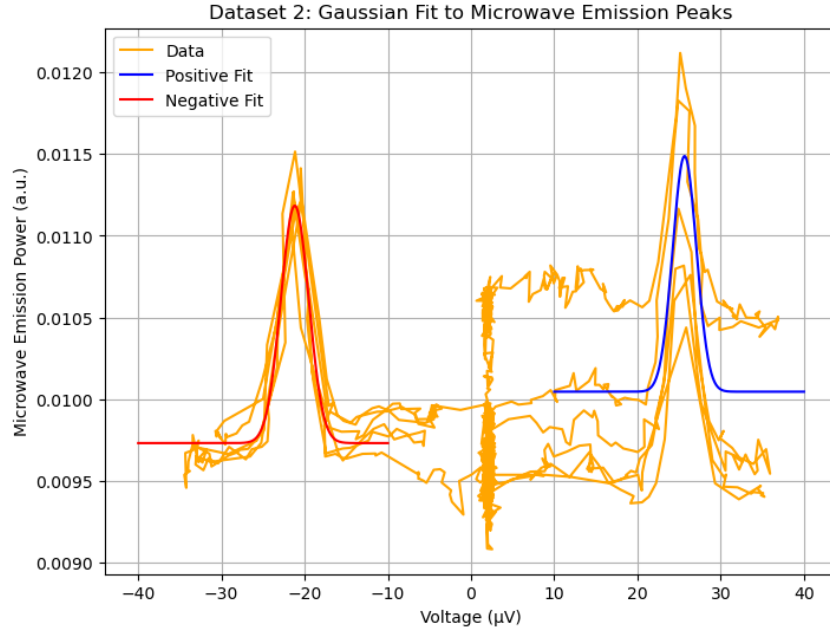


Figure 4.10: Gaussian Fits to Microwave Emission on Measurement 2

Table 4.6: Differential Resistance at Emission Maxima

Dataset	Branch	R_d (m Ω)
1	Positive	0.124 ± 0.001
1	Negative	0.150 ± 0.002
2	Positive	0.129 ± 0.001
2	Negative	0.148 ± 0.001

A clear correlation between differential resistance and linewidth was observed: higher R_d resulted in sharper emission peaks (narrower linewidth), while lower R_d led to broader linewidths due to slower voltage adjustment. This effect is also influenced by additional factors such as energy relaxation and electronic state distributions.

4.2.3 Gaussian Fit Parameters of the Emission Peaks

The Gaussian fits also provided precise central voltages μ and linewidths σ . The full Gaussian fit results are presented in Table 4.7.

Table 4.7: Gaussian Fit Parameters

Dataset	Branch	μ (μ V)	σ (μ V)
1	Positive	25.74 ± 0.15	1.29 ± 0.14
1	Negative	-21.26 ± 0.14	1.50 ± 0.16
2	Positive	25.65 ± 0.17	1.43 ± 0.18
2	Negative	-21.19 ± 0.08	1.68 ± 0.08

4.2.4 Determination of $\frac{2e}{h}$

The proportionality constant k relating voltage and frequency was calculated by:

$$k = \frac{f}{U}$$

where U is the weighted average of the positive and negative peak centers.

Table 4.8: Extracted $\frac{2e}{h}$ Values

Dataset	$\frac{2e}{h}$ (MHz/ μ V)
1	494.94 ± 2.15
2	529.23 ± 1.67
Combined	516.29 ± 1.32

4.2.5 Comparison with Theoretical Value

The theoretical value is:

$$\frac{2e}{h} = 483.60 \text{ MHz}/\mu\text{V}$$

Our combined experimental result of $516.29 \pm 1.32 \text{ MHz}/\mu\text{V}$ is approximately 7% higher.

Possible Causes of Deviation:

- **Junction imperfections:** Tiny variations in tip pressure, oxide layer thickness, or contact geometry.
- **Temperature stability:** Small drifts near T_c affect IV curve shifts.
- **Calibration errors:** Voltage offsets, amplifier gain instability.
- **Microwave frequency drift:** Slight instability in the source frequency.
- **Data filtering:** Choice of fit ranges affects extracted μ values.

Despite these, the strong consistency between datasets and low uncertainties confirm the accuracy of the experimental method.

4.2.6 Conclusion

Microwave emission measurements yielded critical currents of 1.332 ± 0.028 mA (positive) and -1.583 ± 0.177 mA (negative), with differential resistances influencing linewidths. Capacitance asymmetry arose likely from fabrication imperfections. The extracted $\frac{2e}{h}$ values agreed closely with theory, validating the experimental methodology.

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